Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/14/2025
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Single Cell Analysis of Bone Marrow Progenitor and Differentiated Progeny Populations in Response to Long-Duration Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology NNH18ZTT002N:Russian Bion-M2 Mission
Start Date:	06/01/2021	**End Date:**	05/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	Bion-M2		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0929		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In the grant entitled ‘Single cell analysis of bone marrow progenitor cells and differentiated progeny populations following long-duration spaceflight’ we seek to understand and characterize the microgravity-induced molecular genomic changes that occur at the single cell level in rodents during the Bion M2 spaceflight experiment. This will enable the identification and characterization of individual cell and molecular responses to the spaceflight environment and provide insight into the molecular basis for the observed shifts in cellular phenotypes. We specifically hypothesize that gene expression changes induced in bone marrow cells by spaceflight are not gradual and continuous, but instead are the result of precise gene expression state changes in individual cells, and therefore, the identification of these changes will enable the elucidation of underlying regulatory mechanisms. To test our hypothesis, we propose to collect bone marrow cells, spleen, and liver from mice flown on the BionM2 capsule with the following Specific Aims: Specific Aim 1: To determine the transcriptional changes that occur at the single cell level in bone marrow stem cells and differentiated progeny populations. Working Hypothesis: Microgravity and continuous low dose radiation associated with spaceflight causes changes to single cell transcriptome expression profiles leading to repression of differentiation related genes and activation of stem cell proliferation. Specific Aim 2: To compare the effects of microgravity and re-adaptation/recovery on hematopoietic and mesenchymal stem cell lineages in weight bearing versus non-weight bearing bones. Working Hypothesis: Microgravity exposure alters single cell transcriptome expression profiles leading to maintenance of stem cell pluripotency in weight bearing bones (femur/pelvis/tibia), but not in non-weight bearing bones (humerus). Additionally, increased differentiation may be seen in bones of the cranium as a result of increased intracranial pressure and fluid shifts. Specific Aim 3: To determine if spaceflight stresses cause alteration to hematopoietic stem cell (HSC) niche populations in the bone marrow compartment eliciting HSC and differentiated progeny to redistribute to the spleen and liver following long-duration spaceflight exposure. Working hypothesis: Changes in single cell transcriptome expression profiles in marrow stem cells in response to microgravity results in hematopoietic migration to blood forming organs, such as the spleen and liver. Our team at Rensselaer Polytechnic Institute (RPI) has extensive experience in spaceflight research with both in vitro and in vivo cell, tissue, and animal models, including post-flight tissue collection at the Institute of Biomedical Problems (IMBP) in Moscow during the BionM1 flight. Additionally, RPI has recently added several next generation molecular sequencing and library preparation devices and expert users, providing access to a genomics core facility. In summary, we propose the use of revolutionary single-cell transcriptome sequencing technology to characterize the effects of spaceflight on bone marrow progenitor cells and their differentiated progeny. Like other single cell transcriptome studies emerging in the past few years, moving from averaged whole tissue expression profiling to single cell resolution transcriptomics will allow us to attribute specific changes in gene expression levels and gene regulation to individual cell populations. Consequently, this may potentially yield unprecedented insight into the effects of mechanical loading and low dose radiation on progenitor populations and further insight into the ability of both progenitor and differentiated progeny cells to recover from the effects of spaceflight over time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This grant was cancelled. [NOTE: Per the NASA Space Biology Program, these investigations were discontinued (Ed., 5/14/25).]
Bibliography Type:	Description: (Last Updated: 04/03/2025)